

$$X = \begin{bmatrix} 1 & 5 & 10 & 120 \\ 1 & 7 & 12 & 150 \\ 1 & 9 & 14 & 180 \\ 1 & 11 & 16 & 210 \\ 1 & 13 & 18 & 280 \end{bmatrix}$$

$$Y = \begin{bmatrix} 45 \\ 70 \\ 102 \\ 138 \\ 180 \end{bmatrix}$$

$$Q = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 5 & 45 & 70 & 900 \\ 45 & 445 & 670 & 8850 \\ 70 & 670 & 1020 & 13300 \\ 900 & 8850 & 13300 & 179100 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 535 \\ 5741 \\ 8426 \\ 107820 \end{bmatrix}$$

$$Q = \begin{bmatrix} -95.7256 \\ 0.0744 \\ 0.0744 \\ 1.1107 \end{bmatrix}$$

Coefficient of Wind Speed: 0.0744

Coefficient of Bbde Angle: 0.0744

Coefficient of Rotor Speed: 1.167

Intercept : -95.7255

Regression Equation: $0.0744x_1 + 0.0744x_2 + 1.167x_3 - 95.7255$

Power output for windspeed = 20
bbde angle = 18
rotor speed = 290

Power Output: 230.95 Kw

FileEditSelectionViewGoRunTerminalHelp

AL_ML_INTERNSHIP

EXPLORER

AL_ML_INTERNSHIP

- Day1
 - code_1.py
 - Graph.png
 - Load_Extension.csv
 - Report.pdf
- Day2
 - code_2.py
 - power_data.csv
 - Report2.pdf

OUTLINE

- FILESAVED

power_data.csvcode_2.py

Day2 > code_2.py > ...

```
3
4 wind_speed = float(input("Enter Wind Speed: "))
5 blade_angle = float(input("Enter Blade Angle: "))
6 rotor_speed = float(input("Enter Rotor Speed: "))
7 new_data = [[wind_speed, blade_angle, rotor_speed]]
8
9 data = pd.read_csv("Day2/power_data.csv")
10 x = data[["Wind_Speed", "Blade_Angle", "Rotor_Speed"]]
11 y = data["Power_Output"]
12
13 model = LinearRegression()
14 model.fit(x, y)
15
16 slope = model.coef_
17 intercept = model.intercept_
18
19 print(f"Regression Equation: y =(slope)x + {intercept} ")
20
21
22 #predicted = model.predict(pd.DataFrame({'wind_speed': new_data[0], 'blade_angle': new_data[1], 'rotor_speed': new_data[2]}))
23 predicted = model.predict(new_data)
24 intercept_print = -1*intercept #to print the regression equation without plus and minus sign .
25
26 print("Coefficient for Wind Speed:", slope[0])
27 print("Coefficient for Blade Angle:", slope[1])
28 print("Coefficient for Rotor Speed:", slope[2])
29 print("Intercept of the regression line:", intercept)
30 print(f"Regression Equation: {slope[0]:.4f}*x_1 + {slope[1]:.4f}*x_2 + {slope[2]:.4f}*x_3 - {intercept_print:.4f}")
31 print("Predicted power output for the given data is :", predicted[0], 'kW')

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Enter Wind Speed: 20
Enter Blade Angle: 18
Enter Rotor Speed: 290
Regression Equation: y =[0.07444934 0.07444934 1.11674009]x + -95.7255066079298
C:\Users\Aleni\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(
Coefficient for Wind Speed: 0.07444933920704845
Coefficient for Blade Angle: 0.07444933920704844
Coefficient for Rotor Speed: 1.116740088105727
Intercept of the regression line: -95.7255066079298
Regression Equation: 0.0744*x_1 + 0.0744*x_2 + 1.1167*x_3 - 95.7256
Predicted power output for the given data is : 230.95814977973572 kW
PS C:\Users\Aleni\OneDrive\Documents\AI_ML_INTERNSHIP>

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.
+ code_2.py
Describe what to build
+ Auto
Local Default Approvals

Ln 5, Col 50 Spaces: 4 UTF-8 CRLF Python Python 3.12 (64-bit)